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Application Credentials
Ziyi Chen
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Recommendation from Prof. Yi Zhou	Confidential Letter of Recommendation or Evaluation	CONFIDENTIAL
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November 4th, 2024

To Whom It May Concern:

This letter is to express my strongest support for Dr. Ziyi Chen's application of an Assistant Professor position in your department. Ziyi is my first PhD student graduated in 2023, and I am very proud of his academic achievements. He is an outstanding researcher with excellent technical skills in machine learning, optimization, mathematics, and statistics. From the very beginning of his PhD study, Ziyi has demonstrated independence and creativity in identifying and solving research problems, which I have rarely seen from his peers. As a starting topic of his PhD research, Ziyi investigated nonconvex minimax optimization algorithms, which are widely used to enhance the robustness and fairness of machine learning models. In his PhD study, Ziyi focused on studying the emerging multi-agent reinforcement learning problems, and developed efficient learning algorithms with convergence guarantee for these problems in various challenging settings. Ziyi also investigated a broad spectrum of nonconvex optimization problems in machine learning, including federated learning, second-order algorithms for escaping saddle points, bi-level optimization and generalized-smooth optimization. In these directions, Ziyi has achieved results well exceeding the state-of-the-art. Ziyi joined The University of Maryland as a postdoc fellow in 2023, after which he expanded his research interest to large language model alignment using reinforcement learning with human feedback. Throughout his career, Ziyi has emerged as a leading junior researcher in his field, with 15+ papers being published in the top conferences in machine learning including NeurIPS, ICML, ICLR, UAI, etc. He has also published papers in the prestigious Journal of Machine Learning Research (JMLR) and Transaction on Machine Learning Research (TMLR). His research accomplishments have clearly demonstrated his deep insight into the broad machine learning areas and the solid command of a wide-spectrum of statistical and mathematical methodologies. Below, I talk about some of Ziyi's work in more detail.

Multi-agent reinforcement learning (MARL) is an important learning framework for AI agents to learn to make decision by cooperating or competing with each other and also interacting with a dynamic environment. However, these complex interactions bring forth significant challenges, such as non-stationary local environments, high costs of data collection and communication, and the need for privacy protection. Despite these challenges, Ziyi achieved exceptional theoretical and experimental results that far exceeded my expectations. Specifically, in our first study of cooperative MARL where agents collaborate with each other to maximize their average cumulative reward, Ziyi proposed the first fully decentralized actor-critic and natural actor-critic algorithms that not only protect agents' privacy, but also achieve state-of-the-art finite-time sample and communication complexity results. He used multiple novel designs in the algorithms, including noisy local rewards to protect privacy, mini-batch updates to ensure learning accuracy, and fully decentralized Markovian SGD to estimate local natural gradients. With the support of such solid theoretical results, his experimental results demonstrated that the proposed algorithms significantly outperform the existing algorithms. It is worth mentioning that this was his first time to learn and explore policy optimization problem in cooperative MARL. At the beginning, we were not sure whether the idea would work. After just one week, he designed the first efficient and privacy-preserving actor-critic algorithm, derived its complete complexity results, and presented the main algorithm and proof strategies clearly in our weekly meeting – demonstrating his remarkable learning ability. He then showed perseverance in developing the more challenging natural actor-critic algorithm, refining it through multiple versions until it met all desired properties.

Throughout this process, he overcame numerous technical challenges and never backed down. This result is published at ICML 2022.

Another emerging MARL topic that we explored was constrained Markov game, where agents not only compete with each other but also obey behavioral constraints to ensure safety requirements. Through comprehensive literature search, Ziyi found that all the existing works on constrained Markov game aim to find a Nash equilibrium (NE), which is a well-known PPAD-complete problem without polynomial time approach. Then, Ziyi made innovative breakthroughs, including defining correlated equilibrium (CE, a relaxed notion of NE) in constrained Markov game for the first time, designing the first polynomial time approach for constrained Markov game and proving its required polynomial sample complexity to converge to the proposed CE. This result is published at NeurIPS 2022.

In addition to MARL, Ziyi also demonstrates his creative mind and strong mathematical background in other machine learning fields. For example, he has worked on distributionally robust optimization (DRO) problem, an important extension of the classical empirical risk minimization (ERM) approach that is robust to distribution shift. My original expectation was to improve the state-of-the-art complexity result of nonconvex DRO algorithms. However, Ziyi again exceeded my expectation. He proposed an α -symmetric generalized-smooth function class that covers many machine learning applications including nonconvex DRO, phase retrieval, language models, etc., and designed algorithms with near-optimal complexity results for his proposed larger function class, which imply the improved complexity result of nonconvex DRO as desired. This result is published at ICML 2023.

Ziyi is not only an outstanding researcher, but also a valued collaborator. On one hand, he shows respect to others. In particular, he often summarizes other collaborators' constructive feedbacks to make sure he understands them well, and then accordingly improves his ideas, writings, presentations, etc. On the other hand, he contributes many insightful ideas and solid results to our collaborative projects with other graduate students. I am very confident that he is an excellent academic collaborator to comfortably work with.

In 2023, Ziyi joined the Computer Science Department at The University of Maryland as a postdoctoral researcher. Since then, he has independently led two projects on robust reinforcement learning and produced two top-tier conference papers. Now he is actively exploring other emerging areas such as generalization theory on large language model alignment (under review of ICLR 2025) and theoretical foundation of multimodal learning, which are significantly beyond his previous research topics.

In summary, Ziyi is a proliferate researcher. The work he is pursuing will continue to have significant impact on the general theory and algorithms as well as the substantial applications of machine learning. He is absolutely the best and most creative students that I have interacted with in the past as a professor. I believe, without any doubt, that Ziyi will continue his success in his future career.

I thank you very much in advance for your consideration. Should you have any question or require further information, please do not hesitate to contact me.

Sincerely,



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